

EEE3094

Individual Project and Technical Report

Project introduction

School of Engineering • Dr Simon Lambert



What is a final stage dissertation project?

Your stage 3 project is a capstone piece of work which pulls together your theoretical knowledge learned over the course of your degree to produce a body of work which evidences your capability as a professional engineer and demonstrates you are ready to move into the workplace



How much effort should I put into my project?



- Your project is worth 25% of stage 3 – you need to spend a lot of time on the project
 - The MOF states >300 hours
- The rest of stage 3 is worth 75% of the stage – you need to spend three times as much time on the rest of your studies
- You should expect to spend double digit hours on the project every week

Project allocations & supervision

- Your projects have been allocated based on your submitted preferences
- Where projects are over subscribed your stage 2 average mark has been used to rank the preferences
- You will find the list of supervisors and projects on the EEE3094 canvas page
- You will be assigned a project second supervisor in due course
- It is **YOUR** responsibility to make initial contact with your supervisor and meet them in the next ten days.



Project supervision & continuous feedback

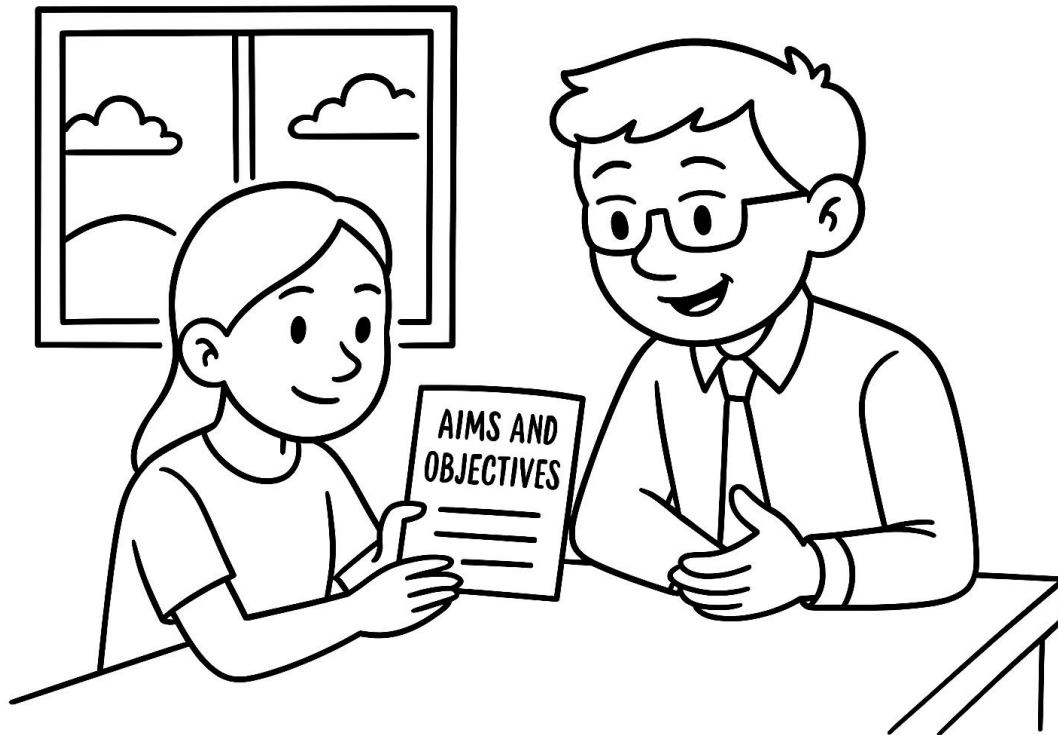
- The relationship with your supervisor will depend on your respective personalities
- You may find it easier to meet in person or online, communicate via email, MS Teams or always in person.
- In all cases, we should always strive to remain professional - If you arrange meetings/calls, please do turn up to them on time and with the agreed information
- Try to set tasks or goals to achieve between meetings and agree them at the time of the meeting recording them in your logbook
 - It's OK to agree tasks for the supervisor (!) such as them providing previous reports or links to specific papers between meetings.
 - Always start a new meeting with a review of the agreed tasks from the last meeting
- Remember, every time you communicate with your supervisor, or anyone else who assists you with your project, you are receiving feedback – make sure you record all feedback in your reflective portfolio and act on it where appropriate
- In the rare case of having issues communicating with your supervisor, please contact me (simon.lambert@ncl.ac.uk)

Individual Project and Technical Report Assessment

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Assessment 1 - Project proposal (Late October, TBC)



- Whilst not formally assessed, this is the written agreement between you and your supervisor on what you propose to undertake in the project
- It includes:
 - a brief introductory narrative including the “problem statement”
 - detailed aims and objectives
 - expected outcomes
 - a detailed project plan in the form of a Gantt chart INCLUDNG gateway milestones and deliverables
- You will return to this plan regularly with your supervisor to review your progress

Assessment 2 – Poster Presentation

- A poster fayre will take place on Friday 13th February 2026.
- Prior to the show you will design and print an academic style poster (i.e. something you might see at a conference) to explain your project and your work to date
- The poster fayre is open to all academics and stage 3 students in EEE
 - You are encouraged to speak to as many people as possible to promote your work
- The poster (and your ability to answer queries) is marked by the second supervisor of your project
- The assessment is worth 15% of the total module mark



Assessment 3 – Project demonstration



- This takes place at the end of your project.
- Depending on the nature of your project you may:
 - Demonstrate your project hardware in the laboratory using appropriate methods/instruments to illustrate its operation
 - Generate a set of simulation files which shows your work in software
- You will be asked about the results you are showing
- The demonstration is marked by both supervisors and is worth 25% of the total project mark

Assessment 4 – Written report (submission May 2026)

- A 6000-word technical report
- This is effectively your “dissertation” submission
- This is a formal piece of work that reflects all of your work on the project.
- It requires you to “own” your project and show their level of skill and knowledge in describing your outcomes.
- You need to demonstrate that you can independently research a technical topic using appropriate resources and report your findings in an effective, professional style
- You are required to assess your own results, reflecting on how they could be made differently and discussing reasons for this, thus showing a deep understanding of your work.
- The report must also demonstrate the project management aspects of the task building on the organic project plan document and should include reflective reference to personal performance against the plan



Project and PECs

- Please note you are not able to self certify for submission of final stage projects
 - i.e. you cannot “save up” your self certifications for the capstone project!
- If you do have a personal extenuating circumstance which affects your project, you should submit this in the usual PEC submission process **BEFORE** the deadline affected.

Project and professionalism

You are expected to treat this project as a professional engineer

- You should keep a dated logbook of your project activity (the session next week will go into details about this)
- It is your responsibility to arrange regular meetings with your supervisor
- You should keep minutes/notes of your meetings with your supervisor and any other interactions you have (e.g. with technicians) in your logbook
- A reflective portfolio is an excellent way to track your progress, record verbal feedback and highlight issues you may want to bring up with your supervisor

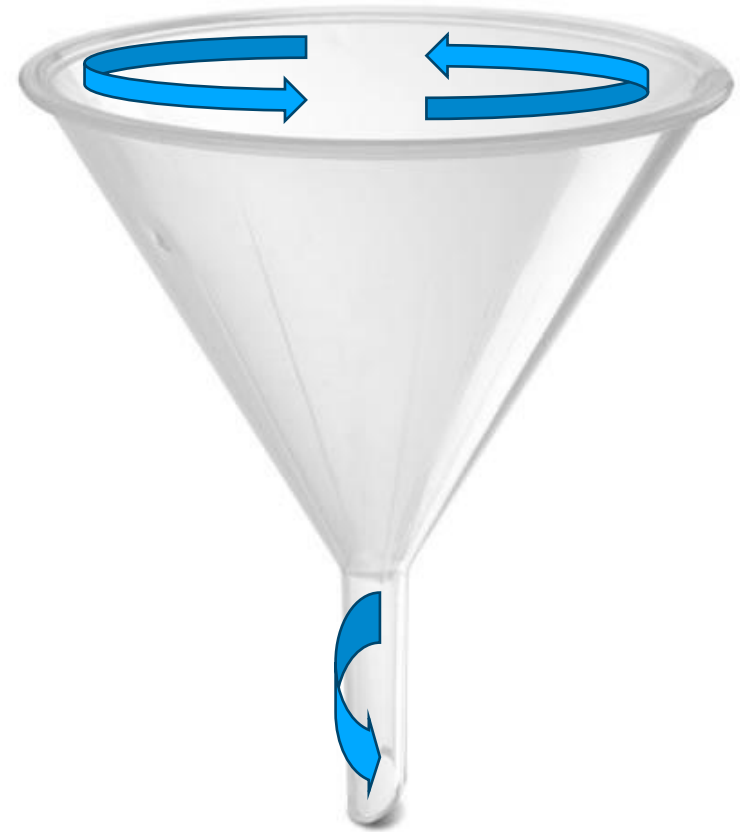
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Project management - 1. Subtasks and what to expect

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Breadth and depth of understanding – the funnel

- Your journey through a research project is like a funnel.
- At the beginning you have to research a wider range of topics in order to contextualise the project
- Towards the end, your breadth of knowledge is much narrower, but considerably deeper!



What does a final stage dissertation project involve?

- Each project is individual and whilst there are similarities in the approach, the exact experience will vary between students and projects.
- Your project supervisor has designed the project around their area of expertise. They will guide you through the nuances of the topic
- In general, the project consists of a series of tasks which can be summarised as:
 - Problem identification and solution scoping
 - Identification of project aims, objectives and specification
 - Solution research (often called “literature review”)
 - Solution design and realisation
 - Solution testing and design iterations
 - Results write-up and conclusions
 - Presentation of findings
- The process is iterative, and you will find yourself returning to the tasks to improve them

Problem identification and solution scoping

- Your supervisor will have written a brief project description in the catalogue
- Hopefully, you will have done some basic research on the topic before selecting it
- You need to meet your supervisor for them to explain the problem in more depth
- Following this initial discussion, you should begin researching the scope of the problem and what you might reasonably expect to achieve in within the project
 - Use appropriate research sources to do this, your supervisor can help guide you. In the first instance it is OK to start with AI, but you should quickly move to more focused, technically appropriate search techniques (see the canvas site)
- The ultimate output of this work is a *problem statement* – i.e. what are you trying to solve/demonstrate within your project.
 - This should cover the problem under study including contextualisation.
 - It should also state the scope with which you will work within (e.g. you will not be building a full-scale EV car charger, but will have practical work limited to 10% sizing, or your project will be completed in the simulation environment etc.) – this is not the final specification!

Identification of project aims, objectives and specification

This process formalises your problem statement with a proposed solution

- Aims:
 - High-level, broad intentions of the project.
 - They describe why the project exists and what it ultimately wants to accomplish.
 - Usually general and abstract, not easily measurable.
- Objectives:
 - Specific, concrete steps that break down the aim into achievable targets.
 - They describe what the project needs to deliver to reach the aim.
 - Use SMART objectives (Specific, Measurable, Achievable, Relevant, Time-bound).
- Specification
 - A detailed description of the requirements — the "how" and "what exactly" the project will deliver.
 - Includes technical details, features, constraints, standards, and acceptance criteria.
 - Provides the blueprint for implementation.

Solution research/literature review

- A literature review can be considered “the intellectual foundation of your project”
- In-depth study of the problem to identify solutions
 - This includes attempts by to solve it (or similar problems).
 - Appraising different approaches by critical analysis both qualitatively *and* quantitatively
 - Identify gaps in the existing work, is there somewhere your work can contribute new knowledge?
- Information sorting is the key skill in this exercise – can you
 - recognise appropriate information sources to your project?
 - demonstrate the relevance to your project?
 - This may include partially attempting to repeat someone else’s results

Solution design and realisation

- Depending on the nature of your project, this may be
 - a series of simulation models being developed
 - a practical hardware build
 - Coding of firmware, a combined hardware/software build
 - All of the above!
- This is where you do the engineering design work in your project
 - You need to ...
 - demonstrate appropriate design steps and decisions
 - plan a suitable approach so subsystem design and testing
 - iterate and change things based on partial results

Solution testing and design iterations

- Here you will test your system, usually in subsystems one at a time
- If the system doesn't work (it won't first time) then you need to demonstrate suitable iterations
- You need to use appropriate methods to test your solutions
 - Are you using the correct equipment?
 - Are the settings correct?
 - Are you able to post-process your results and present them in a user-friendly way
 - Do your results make sense?!

Results write-up and conclusions

- It is extremely common to make a mistake with this section
- Do not leave it too late to write up your results
- Make sure you post-process your results properly and professionally present them
 - Screen grabs from simulation software are not appropriate!!!
 - Charts should be legible and properly labelled
 - Selectively choose the data you want to present (maybe not the whole scope trace etc.)
 - Use the best practice from scientific literature to inform you how to present
- Your conclusions is not a summary
 - Make sure to include informed opinions about the results, why did something happen? What could change it? What would useful future work be?

Presentation of findings

- Finally, you need to demonstrate your project
- This will be your chance to show your system to your supervisors
- Carefully select the experiment(s) you wish to demonstrate